Lab Exercise 17- Creating Secrets in Kubernetes

Name:- Vansh Bhatt

SapID:- 500125395

R.NO:- R2142231689

Batch:- DevOps - B1

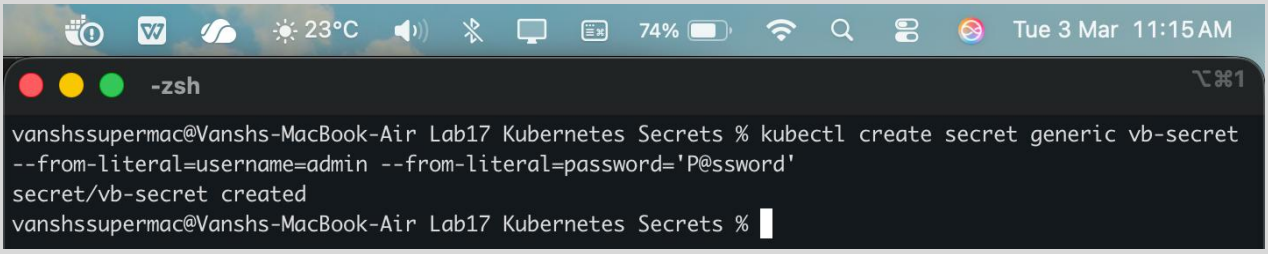
To:- Hitesh Sharma Sir

Creating secrets in Kubernetes is an essential practice for managing sensitive information such as passwords, OAuth tokens, and SSH keys. Let's go through a lab exercise that demonstrates how to create different types of secrets in Kubernetes.

Step 1: Create a Generic Secret

Create a secret from literal values:

```
kubectl create secret generic my-secret --from-literal=username=admin --from-literal=password='P@ssword'
```



Verify the secret:

```
kubectl get secrets my-secret -o yaml
```

```
vanshssupermac@Vanshs-MacBook-Air Lab17 Kubernetes Secrets % kubectl get secrets vb-secret -o yaml
apiVersion: v1
data:
  password: UEBzc3dvcmQ=
  username: YWRtaW4=
kind: Secret
metadata:
  creationTimestamp: "2026-03-03T05:45:03Z"
  name: vb-secret
  namespace: default
  resourceVersion: "18969"
  uid: 9623aa12-dea5-4476-8288-c83be5da1dc7
type: Opaque
vanshssupermac@Vanshs-MacBook-Air Lab17 Kubernetes Secrets %
```

The output will be base64 encoded. To decode, you can use:

echo -n “P@ssword” | base64

```
vanshssupermac@Vanshs-MacBook-Air Lab17 Kubernetes Secrets % echo -n "P@ssword" | base64
UEBzc3dvcmQ=
vanshssupermac@Vanshs-MacBook-Air Lab17 Kubernetes Secrets %
```

Step 2: Create a Secret from a File

Create a file with your sensitive data:

echo -n 'my-username' > ./username.txt

echo -n 'my-password' > ./password.txt

```
vanshssupermac@Vanshs-MacBook-Air Lab17 Kubernetes Secrets % echo -n 'my-username' > ./username.txt
vanshssupermac@Vanshs-MacBook-Air Lab17 Kubernetes Secrets % echo -n 'my-password' > ./password.txt
vanshssupermac@Vanshs-MacBook-Air Lab17 Kubernetes Secrets %
```

Create a secret from the file:

```
kubectl create secret generic my-file-secret --from-file=username=./username.txt --from-file=password=./password.txt
```

```
vanshssupermac@Vanshs-MacBook-Air Lab17 Kubernetes Secrets % kubectl create secret generic my-file-secret --from-file=username=./username.txt --from-file=password=./password.txt
secret/my-file-secret created
vanshssupermac@Vanshs-MacBook-Air Lab17 Kubernetes Secrets %
```

Verify the secret:

```
kubectl get secrets my-file-secret -o yaml
```

```
vanshssupermac@Vanshs-MacBook-Air Lab17 Kubernetes Secrets % kubectl get secrets my-file-secret -o yaml
apiVersion: v1
data:
  password: bXktcGFzc3dvcmQ=
  username: bXktdXNlcm5hbWU=
kind: Secret
metadata:
  creationTimestamp: "2026-03-03T05:49:40Z"
  name: my-file-secret
  namespace: default
  resourceVersion: "19204"
  uid: 667c4db8-2034-4b7a-9ba4-b0f3a0770e22
type: Opaque
vanshssupermac@Vanshs-MacBook-Air Lab17 Kubernetes Secrets %
```

Step 3: Create a Docker Registry Secret

Create a Docker registry secret:

```
kubectl create secret docker-registry my-registry-secret --docker-username=<your-username> --docker-password=<your-password> --docker-email=<your-email> --docker-server=<your-docker-
```

server>

```
ttok@gmail.com --docker-server=https://index.docker.io/v1/
secret/my-registry-secret created
vanshssupermac@Vanshs-MacBook-Air Lab17 Kubernetes Secrets %
```

Verify the secret:

kubectl get secrets my-registry-secret -o yaml

```
24°C  71%  Tue 3 Mar 11:30 AM
-zsh
vanshssupermac@Vanshs-MacBook-Air Lab17 Kubernetes Secrets % kubectl get secrets my-registry-secret -
o yaml
apiVersion: v1
data:
  .dockerconfigjson: eyJhdXRocyI6eyJodHRwczovL2luZGV4LmRvY2tlci5pby92MS8iOnsidXNlcm5hbWUiOiJ2YW5zaGJo
YXR0b2siLCJwYXNzd29yZCI6InZhbnNoRE9DS0VSIiwiZW1haWwiOiJ2YW5zaGJoYXR0b2tAZ21haWwuY29tIiwiYXV0aCI6ImRtR
nVjMmhpYUdGMGRHOXJPblpoYm5Ob1JFOURTMFZTIn19fQ==
kind: Secret
metadata:
  creationTimestamp: "2026-03-03T05:59:05Z"
  name: my-registry-secret
  namespace: default
  resourceVersion: "19660"
  uid: 713561a2-01dd-4130-922b-b7c73e8fd04b
type: kubernetes.io/dockerconfigjson
vanshssupermac@Vanshs-MacBook-Air Lab17 Kubernetes Secrets %
```

Step 4: Create a Secret Using a YAML Manifest

Create a YAML file for the secret:

Create a file named **my-secret.yaml** with the following content:

```
apiVersion: v1
kind: Secret
metadata:
  name: my-manifest-secret
```

```
type: Opaque
data:
  username: YWRtaW4=    # base64 encoded value of 'admin'
  password: UEBzc3cwcmQ=  # base64 encoded value of 'P@ssw0rd'
```

```
apiVersion: v1
kind: Secret
metadata:
  name: mu-manifest-secret
type: Opaque
data:
  username: YWRtaW4=
  password: UEBzc3cwcmQ=

~
~
```

Apply the YAML file:

```
kubectl apply -f my-manifest-secret.yaml
```

```
vanshssupermac@Vanshs-MacBook-Air Lab17 Kubernetes Secrets % vi vb-secret.yaml
vanshssupermac@Vanshs-MacBook-Air Lab17 Kubernetes Secrets % kubectl apply -f vb-secret.yaml
secret/mu-manifest-secret created
vanshssupermac@Vanshs-MacBook-Air Lab17 Kubernetes Secrets %
```

Verify the secret:

```
kubectl get secrets my-manifest-secret -o yaml
```

```
vanshssupermac@Vanshs-MacBook-Air Lab17 Kubernetes Secrets % kubectl get secrets vb-secret -o yaml
apiVersion: v1
data:
  password: UEBzc3dvcmQ=
  username: YWRtaW4=
kind: Secret
metadata:
  creationTimestamp: "2026-03-03T05:45:03Z"
  name: vb-secret
  namespace: default
  resourceVersion: "18969"
  uid: 9623aa12-dea5-4476-8288-c83be5da1dc7
type: Opaque
vanshssupermac@Vanshs-MacBook-Air Lab17 Kubernetes Secrets %
```

Step 5: Use Secrets in Pods

Create a pod that uses the secret:

Create a file named **pod-with-secret.yaml** with the following content:

```
apiVersion: v1
kind: Pod
metadata:
  name: pod-with-secret
spec:
  containers:
  - name: mycontainer
    image: nginx
    env:
    - name: USERNAME
      valueFrom:
        secretKeyRef:
          name: my-secret
          key: username
    - name: PASSWORD
      valueFrom:
        secretKeyRef:
          name: my-secret
          key: password
```

```
apiVersion: v1
kind: Pod
metadata:
  name: pod-with-secret
spec:
  containers:
  - name: mycontainer
   image: nginx
   env:
   - name: USERNAME
    valueFrom:
      secretKeyRef:
        name: my-secret
        key: username
    - name: PASSWORD
     valueFrom:
      secretKeyRef:
        name: my-secret
        key: password
~
~
```

Apply the YAML file:

kubectl apply -f pod-with-secret.yaml

```
vanshssupermac@Vanshs-MacBook-Air Lab17 Kubernetes Secrets % kubectl apply -f pod-with-secret.yaml
pod/pod-with-secret created
vanshssupermac@Vanshs-MacBook-Air Lab17 Kubernetes Secrets %
```

Verify the pod is running and the secret is being used:

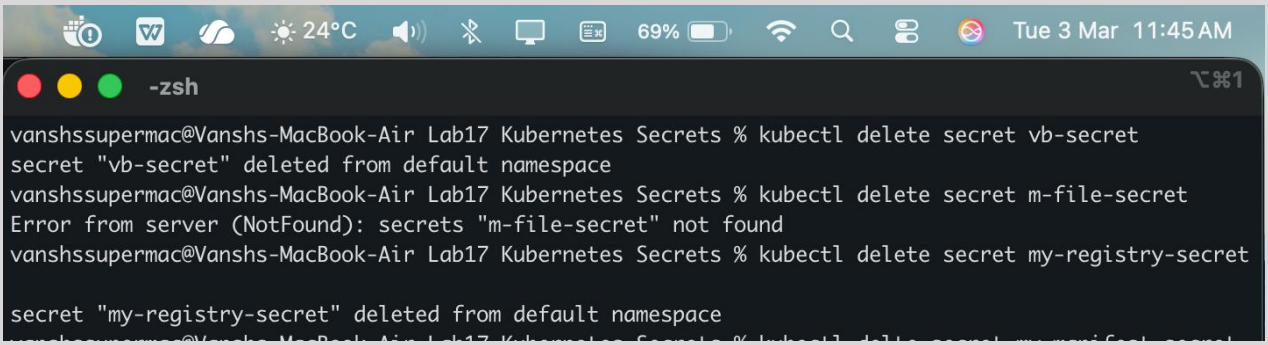
kubectl exec pod-with-secret -- printenv | grep -E 'USERNAME|PASSWORD'

```
vanshssupermac@Vanshs-MacBook-Air Lab17 Kubernetes Secrets % kubectl exec pod-with-secret -- printenv
 | grep -E 'USERNAME|PASSWORD'
```

Step 7: Clean Up

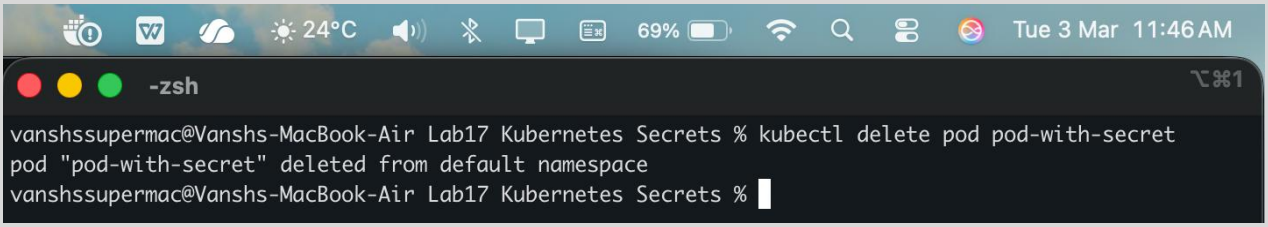
Delete the secrets:

```
kubectl delete secret my-secret
kubectl delete secret my-file-secret
kubectl delete secret my-registry-secret
kubectl delete secret my-manifest-secret
```



Delete the pod:

```
kubectl delete pod pod-with-secret
```



Thank You